Sabir OUSSAOUI, PhD Researcher

University Mohammed 6 Polytechnic UM6P-IWRI
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Languages: English, French, Tamazight C Arabic
Citizenship: Moroccan

PROFESSIONAL EXPERIENCE

Internship: Spatial Assessment of Drought Impact on Farming Systems — Feb 2024 – Sep 2024
University Mohammed 6 Polytechnic (UM6P)/ International Water Research Institute (IWRI)

- *Preprocessed long-time series satellite imagery from multiple platforms.*
- *Cleaned and filtered hydro-climatic data, ensuring accuracy through gap filling and outlier detection.*
- *Extracted phenological parameters and developed spatial-temporal drought maps.*
- *Generated drought metrics using ERA5-Land, MODIS, and Sentinel-2 datasets.*
- *Mapped drought severity impact on arboriculture systems over Tadla and Lower Tassaout plains in Morocco using Sentinel-2 data and machine learning approaches.*

Freelance Trainer – GIS & Remote Sensing — Dec 2023 – Mar 2024
SIGOBEL Center, Morocco. Remote

- *Conducted online training sessions, instructing over 200 individuals in GIS and remote sensing.*
- *Delivered hands-on courses using Google Earth Engine.*
- *Processed and filtered raster and vector data for analysis.*
- *Calculated various spectral indices for land cover assessment.*
- *Worked with digital terrain models (e.g., shading, slope analysis).*
- *Created land cover classification maps and performed accuracy assessments.*
- *Computed confusion matrices to evaluate classification results.*
- *Produced scientific research maps for GIS applications.*

Field Internship in Geology — Dec 2023 - Dec 2023
Université Sultan Moulay Slimane, Faculty of Science and Technology, Beni Mellal.

- *Conducted a geological study of the Chougrane Graben and the Central High Atlas.*
- *Analyzed and recorded stratigraphic sequences.*
- *Created a small-scale geological map of the Chougrane Graben.*

Internship: Geospatial Database Management — Aug 2023 – Oct 2023
University Mohammed 6 Polytechnic (UM6P)/Center for Remote Sensing Applications (CRSA)

- *Managed geospatial data as part of the MorSnow_2 project.*
- *Worked with GIS tools, including QGIS, PostgreSQL/PostGIS, and GeoServer.*
- *Developed Python scripts in Jupyter Notebook for data processing and analysis.*
- *Gained expertise in geospatial data management and its applications in geography and environmental sciences.*

Internship: GIS & Remote Sensing — Jul 2023 - Oct 2023
Géomatique Métier d'Avenir (Remote)

- *Used Google Earth Engine to analyze land cover dynamics for 1986, 2000, 2020, and 2022.*
- *Created land cover classification maps.*
- *Calculated spectral indices, including albedo, TSGI, NDVI, MDWI, MSAVI, and LST.*

EDUCATION

PhD Student in Climate Change, Remote Sensing and AI (Cotutelle) — Oct 2025-Current
Québec Trois-Rivières University (UQTR)/ Research Centre for Watershed–Aquatic Ecosystem Interactions (RIVE)

- ***Thesis titled**: Drought Monitoring and Forecasting in Morocco using Remote Sensing and Socioeconomic Data*

PhD Student in Climate Change, Remote Sensing and AI — Oct 2024 - Current
University Mohammed 6 Polytechnic (UM6P)/ International Water Research Institute (IWRI)

- ***Thesis titled**: Drought Monitoring and Forecasting in Morocco using Remote Sensing and Socioeconomic Data*

1. **Pre-Doctoral Fellow (PDF)- Drought Monitoring and Forecasting in Morocco using Remote Sensing and Socioeconomic Data,** — Oct 2024-Sep 2025
   - *Conducting a comprehensive literature review to identify research gaps in drought monitoring.*
   - *Leading a field survey of 204 farmers to assess the socioeconomic impact of drought.*
   - *Processing and analysing remote sensing and AI-driven data for drought forecasting.*
   - *Engaged in predoctoral coursework proposed by CAES to enhance expertise.*

Master's degree in environmental Geomatics — Sep 2022 - Jul 2024
Faculty of Sciences and Technics, Sultan Moulay Slimane University (USMS), Beni Mellal, Morocco.

- ***PFE thesis**: Spatial Assessment of the Impact of Drought on Agricultural Systems in the Oum Er Rbia Basin.*

Bachelor's Degree in Geomatics and Territorial Planning — Sep 2021 - Jul 2022
Faculty of Sciences and Technics, Sultan Moulay Slimane University (USMS), Beni Mellal, Morocco.

- ***PFE thesis**: Photogrammetric restitution of the MGHILA Campus using stereoscopic images from the Phantom 4 RTK drone using Agisoft Metashape.*

University Diploma in Sciences and Technics (DEUST) in Biology, Chemistry, And Geology — Sep 2019 - Jul 2021
Faculty of Science and Technics, USMS, Beni Mellal, Morocco.

High School Diploma in Experimental Sciences — Sep 2018 - Jun 2019
Hassan I High School, Taguelft Azilal, Morocco.

- ***Option***: Life and Earth Sciences

REPORT OF TEACHING AND TRAINING

Institute	Program name/degree/discipline	Course	Academic Year
UM6P	Instructor: Tensift Water Basin Agency & Directorate of Water Research and Planning Rabat	Lectures, Tutorials, and Practical Works in Remote Sensing and GIS	06/2025
USMS-FST	Instructor: Master 2 in environmental geomatics	Lectures, Tutorials, and Practical Works in Remote Sensing Using Google Earth Engine	04/2025
USMS-FST	Instructor: Master 1 in environmental geomatics	Lectures, Tutorials, and Practical Works in Remote Sensing Using Google Earth Engine	02/2025

SELECTED MANAGEMENT OF THESIS (PFE, BACHELOR, MASTER)

Student name	Thesis title	Discipline (University)	Academic Year	Degree / Position
Ayoub GOULAHIANE	Spatiotemporal analysis of agricultural drought through land use dynamics and artificial intelligence approaches in the Doukkala plain from 2016 to 2025	USMS- MSc Sciences and Technics, Environmental Geomatics	2025	MSc / external supervisor
Ahmed TIOUAL	The impact of drought on rainfed cropping systems using remote sensing data	UCA - MSc Sciences, Data Science	2025	MSc / co-external supervisor
Maryem ELKHATTAB / Mohammed ELOUAHMANI	Cartographic Analysis of Drought Effects in the Oum Er-Rbia Basin: Application of Spectral Indices and Google Earth Engine (GEE)	USMS- BSc Sciences and Technics, Geomatics and Territory Management	2025	BSc / external supervisor
Fatima OUQELLA / Ahlam EN NOUARY	Geomatics approach to household waste management: study of the city of Rommani	USMS- BSc Sciences and Technics, Geomatics and Territory Management	2025	BSc / co-external supervisor
Hamid AIROU / Ayoub SAAID	Detection and mapping of urban expansion using GIS and spatial remote sensing: the case of the Fquih Ben Salah city	USMS- BSc Sciences and Technics, Geomatics and Territory Management	2025	BSc / external supervisor

SCIENTIFIC & ACADEMIC COMMUNICATIONS

Paper / Article

Nb	Citation	IF/CS	Q
5	Hanadé Hoummad, Ismaguil, Abdessamad Hadri, Abdelghani Boudhar, El Mahdi El Khalki, Ismail Karaoui, Sabir Oussaoui, Samih Mohamed, and Christophe Kinnard. "Development of a Hydrometeorological Drought Severity Composite Index Based on the Integration of Multisource Characteristics and an Explainable Artificial Intelligence Model."	5/7.4	Q1
4	Hanadé Houmma, Ismaguil, Abdessamad Hadri, Abdelghani Boudhar, El Mahdi El Khalki, **Sabir Oussaoui**, Ismail Karaoui, Azzeddine El Midaoui, and Christophe Kinnard. "Temporal relationships between agricultural and meteorological drought over the Oum Er Rbia River, Morrocco." Big Earth Data (2025): 1-36.	4.2/7.2	Q1
3	**Sabir Oussaoui**, Abdelghani Boudhar, Abdessamad Hadri, Youssef Lebrini, Ismaguil Hanadé Houmma, Ismail Karaoui, El Mahdi El Khalki, Jamal-eddine Ouzemou & Christophe Kinnard (**2025**) Mapping drought severity impact on arboriculture systems over Tadla and lower Tassaout plains in Morocco using Sentinel-2 data and machine learning approaches, Geocarto International, 40:1, 2471104, DOI: 10.1080/10106049.2025.2471104	3.3/7.2	Q1
2	Houmma, Ismaguil Hanadé, Abdessamad Hadri, Abdelghani Boudhar, Ismail Karaoui, **Sabir Oussaoui**, El Mahdi El Khalki, Abdelghani Chehbouni, and Christophe Kinnard. "Analysis of the Propagation Characteristics of Meteorological Drought to Hydrological Drought and Their Joint Effects on Low-Flow Drought Variability in the Oum Er Rbia Watershed, Morocco." *Remote Sensing* 17, no. 2 (**2025**): 281.	3.3/5.3	Q1

1	Ouzemou, Jamal-Eddine, El Harti, Abderrazak, El Hachimi, Jaouad, Lhissou, Rachid and **OUSSAOUI, Sabir**, Optimizing Sugar Beet Crop Management: Integrating Satellite Imagery for Precision Agriculture. Smart Agricultural Technology (Accepted with revisions, 2024) Preprint: http://dx.doi.org/10.2139/ssrn.4570968	6.3/4.2	Q1

Scientific Seminars and conferences

Selected International Conferences

6	**Oussaoui, Sabir**, Abdelghani Boudhar, Abdessamad Hadri, Youssef Lebrini, Ismaguil Hanadé Houmma, Ismail Karaoui, El Mahdi El Khalki, Jamal-eddine Ouzemou, and Christophe Kinnard. *Mapping Drought Severity Impact on Arboriculture Systems over Tadla and Lower Tassaout Plains in Morocco Using Sentinel-2 Data and Machine Learning Approaches*. Poster présenté au XIX World Water Congress, International Water Resources Association (IWRA), Marrakech, Maroc, 1–5 décembre 2025.	2025
5	**Oussaoui, Sabir,** Abdelghani Boudhar, Abdessamad Hadri, Youssef Lebrini, Ismaguil Hanadé Houmma, Ismail Karaoui, El Mahdi El Khalki, Jamal-eddine Ouzemou, and Christophe Kinnard. *Remote Sensing and Machine Learning for Drought Severity Assessment in Arboriculture Systems of the Tadla and Lower Tassaout Plains, Morocco*. Poster présenté aux Doctoral Days, UM6P - University Mohammed VI Polytechnic, Ben Guerir, Maroc, 2025.	2025
4	**Oussaoui, Sabir**, Abdelghani Boudhar, Abdessamad Hadri, Youssef Lebrini, Ismaguil Hanadé Houmma, Ismail Karaoui, El Mahdi El Khalki, Jamal-eddine Ouzemou, and Christophe Kinnard. *Mapping Drought Severity Impact on Arboriculture Systems over Tadla and Lower Tassaout Plains in Morocco Using Sentinel-2 Data and Machine Learning Approaches*. Poster présenté à l'**Earth Observation (EO) Workshop on Assessing Climate Impacts Across Africa**, AfEOP Project, UM6P-MIT, 2025.	2025
3	Achemrk A.; El Battay A.; Laamrani A.; Ouzemou J.E.; Chehbouni A.; **Oussaoui S.;** (2025); *Monitoring of soil salinity using 3D feature space models and machine learning algorithms: A case study of Sehb El Msjoune semi-arid and arid area, Morocco*. The Third International Laayoune Forum on Biosaline and Arid Land Agriculture. African Sustainable Agriculture Research Institute (ASARI-UM6P), Laayoune, Morocco	2025
2	**Oussaoui, S.**; Boudhar, A.; Hadri, A.; Lebrini, Y.; Houmma, I.H.; Karaoui, I.; Kinnard, C.; El Khalki, E.M.; Ouzemou, J.E.; El Midaoui, A.; Chehbouni, A. (2024). *Spatial assessment of the impact of drought on agricultural systems in the irrigated perimeter of tadla and tassaout and the response time of arboriculture*. 6th International Association for Hydro-Environment Engineering and Research (IAHR) Africa Congress, Benguerir-Marrakech, Morocco. Mohammed VI Polytechnic University.	2024
1	Houmma, I.H.; Hadri, A.; Boudhar, A.; Karaoui, I.; **Oussaoui, S.;** El Khalki, E.M.; Kinnard, C. (2024). *A new hydrological drought index based on spatio-temporal changes in annual runoff using rainfall-runoff models*. 6th International Association for Hydro-Environment Engineering and Research (IAHR) Africa Congress, Benguerir-Marrakesh, Morocc. Mohammed VI Polytechnic University.	2024

Scientific Seminars & Advanced Training

Program	Institution(s)	Description	Year
Advanced School on High-Performance Computing and Applied AI for High-Resolution Regional Climate Modeling	ICTP (International Centre for Theoretical Physics), UM6P, IAEA, UNESCO	Completed intensive advanced training on high-performance computing (HPC), applied artificial intelligence, and regional climate modeling. Covered numerical modeling, computational methods, and AI	2025

		applications for high-resolution climate simulations. (Ben Guerir, Morocco)	
Pedagogical Field Exercise – Land-Use Diagnostic Study (Tensift Basin)	UM6P – Center for Remote Sensing Applications (CRSA) & AgroParisTech	Selected to participate in a joint UM6P–AgroParisTech applied training program (RUMP8 Ben Guerir). Conducted land-use diagnostic analysis including field investigation, problem definition, methodological design, data analysis, group reporting, and oral presentation. Focused on territorial agricultural assessment in the Tensift watershed.	2025

HONORS & AWARDS

Award	Organization	Description	Year
IWRA Silver Membership Award – Early Career Lightning Talks	International Water Resources Association (IWRA)	Selected as one of the top early-career researchers at the XIX IWRA World Water Congress. Recognition of research on drought monitoring, agricultural resilience, and water sustainability. Includes one-year Silver Membership.	2025
Top 3 Project Award – IWRI Water Day Hackathon	International Water Research Institute (IWRI), UM6P	Awarded among the Top 3 projects for developing an innovative solution integrating remote sensing, environmental intelligence, and data-driven decision support for water challenges.	2025